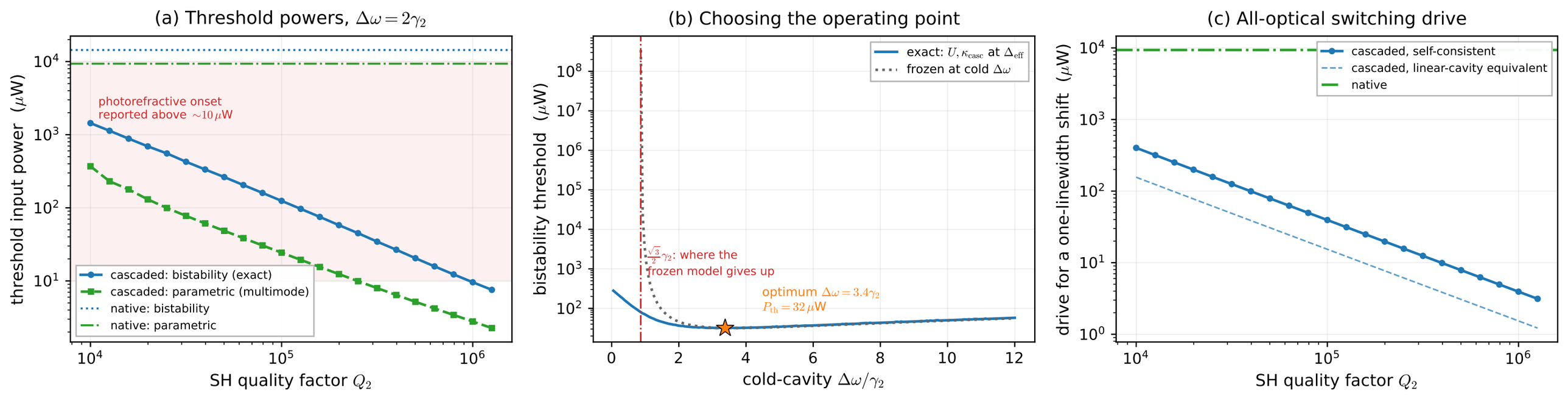
(a) Threshold powers, $\Delta\omega = 2\gamma_2$
threshold input power (μW)
SH quality factor Q_2
photorefractive onset
reported above $\sim 10\,\mu$W
cascaded: bistability (exact)
cascaded: parametric (multimode)
native: bistability
native: parametric
(b) Choosing the operating point
bistability threshold (μW)
cold-cavity $\Delta\omega/\gamma_2$
exact: $U, \kappa_{\mathrm{casc}}$ at Δ_{eff}
frozen at cold $\Delta\omega$
$\frac{\sqrt{3}}{2}\gamma_2$: where the
frozen model gives up
optimum $\Delta\omega = 3.4\gamma_2$
$P_{\mathrm{th}} = 32\,\mu$W
(c) All-optical switching drive
drive for a one-linewidth shift (μW)
SH quality factor Q_2
cascaded, self-consistent
cascaded, linear-cavity equivalent
native